

Replication Report: *Loop Current Order on the Kagome Lattice* arXiv:2506.01648v2 (Zhan et al., 2026)

Textures-100 replication (loop-current class)

July 19, 2026

1 Summary

Zhan *et al.* study the spinless kagome lattice at the p-type Van Hove filling ($\mu = 0$) with nonlocal density–density repulsion (nearest-neighbor V_1 , second-neighbor V_2), Eq. (1):

$$H = -t \sum_{\langle ij \rangle} (c_i^\dagger c_j + \text{h.c.}) - \mu \sum_i n_i + V_1 \sum_{\langle ij \rangle} n_i n_j + V_2 \sum_{\langle\langle ij \rangle\rangle} n_i n_j.$$

Using unbiased functional renormalization group (FRG), they find that sublattice interference suppresses onsite CDW and that a 2×2 **(3Q) loop-current order (LCO)** emerges as the many-body ground state over a significant region of strong V_2 . The resulting electronic model with LCO is a **Chern insulator with total $C = 1$** (quantum anomalous Hall), analogous to the Haldane model.

We reuse the shared Textures-100 loop-current kagome kernel (`loop_current_kagome_kernel.py`; provenance in `PROVENANCE.md`) and add a folded 2×2 tight-binding model carrying an imaginary (loop-current) bond order on 1nn and 2nn bonds. The FRG phase diagram itself is out of scope (HPC method); we replicate the paper’s downstream, machine-checkable single-particle claims.

2 Claims tested

- A** Bare kagome spectrum: flat band, Dirac cone at K , p-type VHS saddle at $E = 0$ ($\mu = 0$).
- B** Imaginary (loop-current) bond order opens a full gap (Fig. 3d).
- C** The gapped LCO state is a Chern insulator, total $C = 1$ (Fig. 3d, $\Delta_{1nn}^{\text{LCO}} = 0.1t$, $\Delta_{2nn}^{\text{LCO}} = 0.15t$).
- D** Landau quartic Eq. (2), $Z_1 - Z_2 > 0$, selects the equal-weight 3Q state.
- E** TRS is broken by the imaginary channel (LCO current $\neq 0$), and the 3Q state carries a net ferromagnetic orbital moment.

3 Results

4 Discussion

All five in-scope claims reproduce. The headline physics — a spinless kagome tight-binding model with a 2×2 imaginary-bond (loop-current) order that gaps the p-type-VHS Fermi surface into

Claim	Result (this replication)	Paper	Verdict
A	$E(\Gamma)=[-4, 2, 2]$, $E(K)=[-1, -1, 2]$, $E(M)=[-2, 0, 2]$; flat band $+2t$ (spread 0); Dirac touch at K ; p-type VHS at $E=0$ (M mid-band = 0, lower-band DOS peak at 0).	flat band, Dirac, p-type VHS at $\mu=0$	PASS
B	Folded 12-band LCO model ($\Delta_{1nn}=0.1t, \Delta_{2nn}=0.15t$): positive indi- rect gap at fillings 4 (0.028), 5 (0.032), 8 (0.030).	full gap around E_F	PASS
C	Chern by gapped filling {4:+1, 5:+1, 8:+2, 10:-1}; total $C = 1$ at the VHS-relevant filling. Kernel cross-check: staggered flux gives $C = -1$ (Haldane/OMN).	total $C = 1$, QAH	PASS
D	$E(1Q)=0.50 > E(2Q)=0.35 > E(3Q)=0.30$; 20k simplex search $\rightarrow$ equal weights (0.58, 0.58, 0.58).	equal-weight 3Q, 2×2	PASS
E	Loop current (imag bond) = -0.195 ; charge (real bond) = $+0.220$ with current ≈ -0.013 ; 3Q (1, 1, 1) dipole = 1 (FM), 2Q (1, 1, 0) dipole = 0.	TRS-broken LCO; 3Q FM ($\sim$ $0.03 \mu_B/\text{site}$ FeGe)	PASS

a $C = 1$ Chern insulator, with the 3Q state selected by the TRS-broken Landau quartic — is recovered quantitatively for the spectrum and qualitatively (with the correct topological invariant) for the many-body-derived electronic model.

The paper’s defining *method* (unbiased FRG phase diagram, Fig. 2) is not reproduced: it requires a dedicated FRG code and HPC. This is documented in `failure_analysis.md`, not obscured. The folded-model current-sense assignment is geometric rather than the paper’s renormalized-susceptibility eigenvector, so higher-filling Chern values differ; the VHS-relevant filling carries $C = 1$ as claimed.

5 Verdict

REPRODUCED (in-scope single-particle claims).

Coverage: 8/10 — all downstream electronic-model claims covered; the FRG many-body phase diagram and SC channels are out of scope.

Agreement: 8/10 — spectrum exact; gap/Chern/Landau/3Q all match with convention-level caveats on gap magnitude and residual bond current.